

Yñikko Arzee Neo Aguas

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SUMMARY

Computer Science student specializing in Artificial Intelligence, with two IEEE-published papers in medical imaging and clinical NLP. Experienced in building high-accuracy computer vision models and Retrieval-Augmented Generation (RAG) systems for healthcare, and in shipping full-stack, production software.

EDUCATION

Mapúa University

Bachelor of Science in Computer Science, Specializing in Artificial Intelligence

Makati City

Aug. 2023 – Oct. 2026

TECHNICAL SKILLS

Languages: Python, SQL (Postgres), JavaScript, TypeScript, HTML/CSS, Java, C++

Frameworks & Libraries: React, Node.js, Next.js, Vite, pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow

AI/ML: Computer Vision, RAG, NLP, YOLO, nnU-NetV2, FAISS, ChromaDB

Developer Tools: Git, GitHub, Docker, Google Colab, Google Cloud Platform, VS Code, IntelliJ

EXPERIENCE

Software Engineer Intern | Exoasia Innovation Hub

Apr. 2026 – Jun. 2026

- Shipped 15+ production feature modules for a 3-role internal platform used by 30+ people, architecting a React + Supabase data layer with role-based access enforced through 25+ PostgreSQL migrations and Row-Level Security. (*React, Supabase, PostgreSQL*)
- Built X-O, an in-app AI assistant with a slash-command palette, and added real-time sync across 10+ modules that cut manual data entry and enabled live updates for announcements, messages, and project changes. (*Supabase Edge Functions, Supabase Realtime*)
- Delivered Google OAuth, a full dark/light theming system, and Excel/PDF export across a responsive UI spanning mobile, tablet, and desktop. (*React, OAuth, TypeScript*)

PUBLICATIONS

RAG-Based Clinical Guideline Chatbot for Atrial Fibrillation | Python, Gradio

Dec. 2025 – Mar. 2026

- Published IEEE paper on a RAG chatbot answering clinical questions grounded in 100+ pages of ESC atrial-fibrillation guidelines, achieving a BERTScore F1 of 0.835.
- Built the retrieval pipeline with biomedical embeddings, vector search, and reranking, raising answer faithfulness to 8.75/10 across 20 clinical queries. (*MedCPT, FAISS, BGE*)
- Deployed quantized LLMs to reach 0.7s latency and 7.9 tokens/sec for real-time clinical decision support. (*Llama-3, Phi-3, Qwen3, Ollama*)

Multi-Class Kidney Abnormality Segmentation | Python, Roboflow, Streamlit

Aug. 2025 – Jan. 2026

- First-authored IEEE paper implementing a YOLOv12 architecture to detect and segment cysts, stones, and tumors in CT scans, achieving mAP@0.5 of 0.946 (axial) and 0.885 (coronal).
- Curated and preprocessed a class-balanced dataset of 14,761 axial and coronal CT scans with high-fidelity clinical annotation. (*Roboflow*)
- Deployed a Streamlit web interface for real-time, privacy-preserving inference and color-coded visualization of abnormalities.

PROJECTS

Edge AI Food Label Analysis | Python, YOLOv8, RapidOCR, RAG, ChromaDB, Raspberry Pi 5

- Built an offline dietary-analysis system deployed entirely on a Raspberry Pi 5, grounding responses in WHO, FDA, and FARE health guidelines with no cloud dependency.
- Engineered a per-field retrieval strategy over a 682-chunk vector store, achieving a mean cosine similarity of 0.854 with all queries clearing the 0.75 confidence threshold.
- Deployed a 4-bit quantized Llama 3.2 3B model via Ollama, cutting the memory footprint from 6GB to 2GB to fit the device's 8GB budget.

- Integrated YOLOv8 label detection (0.969 mAP50) with the language pipeline via a subprocess architecture isolating conflicting ARM dependencies.

Heart Disease Risk Detection | *Python, Streamlit, Scikit-learn*

- Led preprocessing and cleaning of complex medical datasets to prepare them for machine learning pipelines.
- Evaluated Logistic Regression, SVC, and Random Forest models to assess patient cardiovascular risk.
- Tuned hyperparameters with Grid Search CV to improve prediction accuracy.

Earthquake Data Analysis & Prediction Tool | *Python, Streamlit, Scikit-learn, Pandas, Seaborn*

- Developed a multi-page Streamlit web application for end-to-end earthquake analysis, from exploratory data analysis to live prediction.
- Trained Random Forest, KNN, and Linear Regression models for tsunami, alert-level, and magnitude prediction, reaching 96% top accuracy.
- Designed geospatial density maps and KDE/temporal plots, each paired with plain-language interpretations for non-technical readers.

Vault (File Manager) | *React, Supabase, Shadcn UI*

- Architected a file management system with organization, favorites, and a secure trash section, using Shadcn UI for a responsive, accessible interface.
- Built custom account-management modules, backed by Supabase for secure cloud storage.

MyApt (Apartment Dashboard Management System) | *React, Firebase*

- Built a property-management platform with real-time data tracking, automated reporting, and a tenant-manager messaging center.
- Integrated secure authentication and tracked feature development with Jira across mobile and desktop.

CERTIFICATIONS

Google Cybersecurity Professional Certificate (6-course series) — Google

CCNA: Switching, Routing, and Wireless Essentials — Cisco Networking Academy (Oct. 2025)

Data Science & AI: Data Science Methodology (IBM), Big Data & AI Ethics (UC Davis), AI Data Fairness and Bias (LearnQuest)

Software Engineering: Intro to Software Engineering (IBM), Data Structures (UCSD), Programming Languages (Univ. of Washington)

UI/UX Design Specialization — CalArts